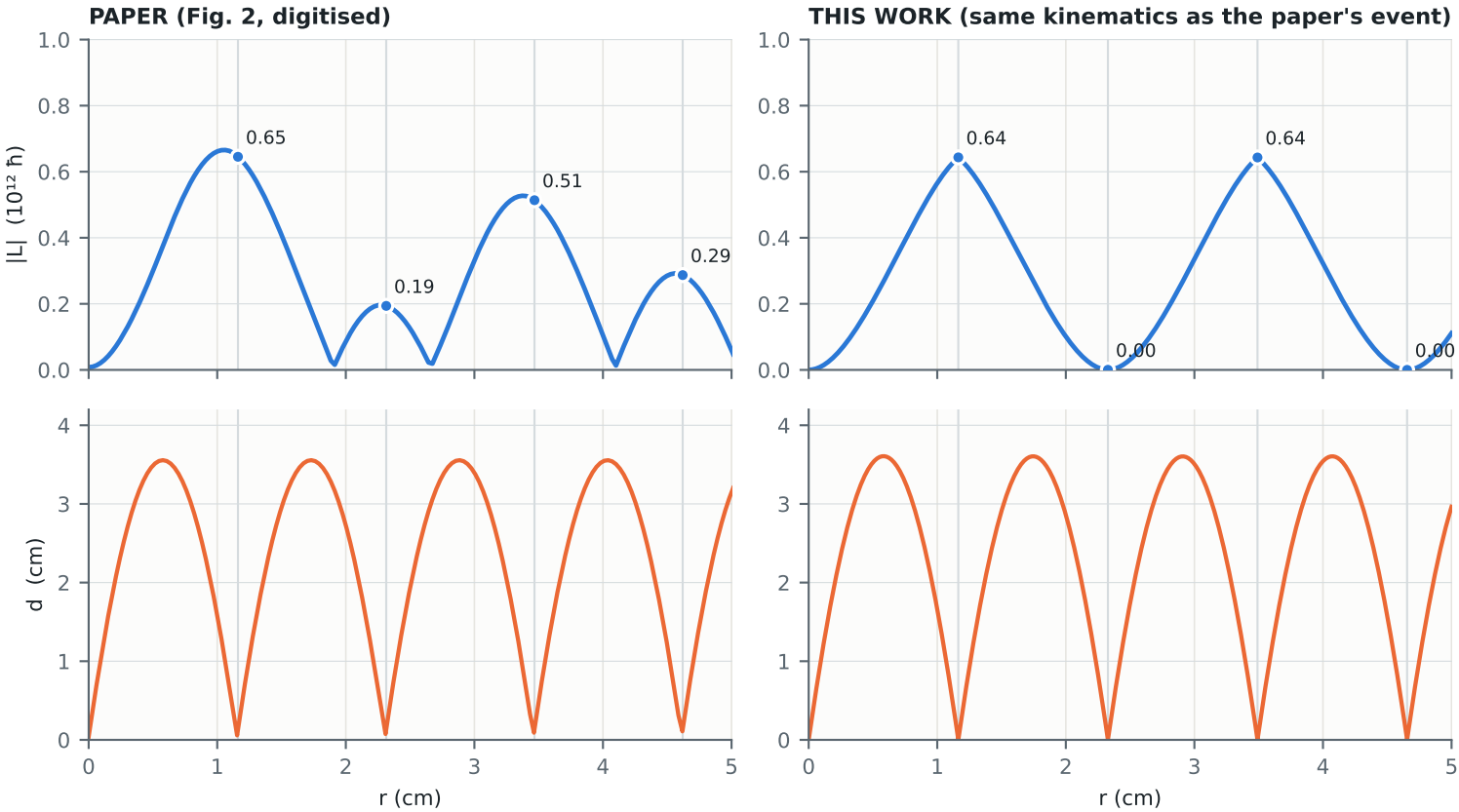
Fig. 2 of arXiv:1708.02243: paper vs this work

Colored quirk, m_Q = 1.8 TeV, Λ = 2 keV, B = 2 T. |L| and quirk separation d vs r (transverse radius of the CM).



Grey lines and dots: closest approaches (minima of d). Our curve: v* = 0.556, β_cm = 0.0975, charges ±2/3, 3000 RK4 steps per period.
The paper does not give the event's initial conditions; v* and β_cm are read from its d curve (d max and spacing).

Values

Quantity	Paper	This work (same kinematics)	This work (simulated event, key 8)
d max (cm)	3.56	3.61	4.19
Spacing of closest approaches in r (cm)	1.153	1.163	1.047
First |L| peak (10¹² ħ)	0.666	0.643	0.862
|L| at closest approaches 1-4 (10¹² ħ)	0.65 / 0.19 / 0.51 / 0.29	0.64 / 0.00 / 0.64 / 0.00	0.86 / 0.00 / 0.86 / 0.00
r of closest approaches (cm)	1.16 / 2.31 / 3.47 / 4.62	1.16 / 2.33 / 3.49 / 4.65	1.05 / 2.09 / 3.14 / 4.19
r where |L| = 0 (cm)	1.91 / 2.66 / 4.10	2.33 / 4.65	2.09 / 4.18
Change of L per half-oscillation (signed, 10¹² ħ)	+0.65 / -0.84 / +0.71 / -0.80	+0.64 / -0.64 / +0.64 / -0.64	±0.86 (all equal)
Charges	not stated	±2/3	±1

Columns: same kinematics = event built from the paper's d curve (v* = 0.556, β_cm = 0.0975), charges ±2/3;
simulated key 8 = event from our MadGraph + Pythia sample with the nearest kinematics, charges ±1.

Matches: d max, spacing (both by construction for the middle column) and the first peak.
Differs: after the first peak our |L| returns to 0 at every second closest approach; the paper's does not (0.19, 0.29).
In the paper the half-oscillations of L are unequal (+0.65 / −0.84 / +0.71 / −0.80); in ours they are equal (±0.64).

Sources: tables/fig2_r_d_L_*.csv, FINAL_FIGURES/F34_fig2_v4; script scripts/R_fig2_sheet_2026-10-12.py

Equations used vs the paper

Paper	What	In our code	Same as paper?
Eq. 3	$\frac{d}{dt}(m\gamma\mathbf{v}) = -\Lambda^2\left(\frac{\sqrt{1-v_\perp^2}}{v_\parallel}\mathbf{v}_\parallel + \frac{v_\parallel}{\sqrt{1-v_\perp^2}}\mathbf{v}_\perp\right) + \mathbf{F}_{\text{ext}}$	Same expression, term by term (RK4). v‖ = v·ŝ (signed, ŝ along the string), so v‖(vector)/v‖ = ŝ	Yes
Eq. 3, F_ext	Lorentz force (+ material, Sec. II)	q v × B, B = 2 T along z; no material	Lorentz: yes. Material: neglected in the paper (Sec. II); none used
Eq. 4	d_cm(t): one quirk's distance from the CM (max (γ−1) m_Q/Λ²)	d = full separation = 2 d_cm d max 3.607 cm vs 2(γ−1)m_Q/Λ² = 3.607	Yes
Eq. 1	Amplitude ~2 cm (γ−1)(m_Q/100 GeV)(keV/Λ)²	1.80 cm per quirk (constructed event)	Yes
Eq. 5	Torque τ ~ 2d × eE_cm, E_cm = γ_cm v_cm × B	Exact CM-frame EM torque = q d × E_cm per half-oscillation (±0.556)	Yes
Eq. 6	L ~ e v_cm γ_cm vγ(γ−1) m_Q² B/Λ⁴	Estimate 0.32×10¹² ħ; trajectory 0.64×10¹² ħ	Same order (Eq. 6 is an estimate)
Fig. 2	L, d vs r of the CM	L = d* × (p1* − p2*)/2 in the CM frame at equal CM time; d = |x1* − x2*|; r = CM radiu	Frame of L: paper does not state; we used CM
—	Integrator, step size	RK4, 3000 steps per oscillation period	Paper does not state

Checks done on our Fig. 2

- L definition: CM frame and three lab-frame definitions give the same |L| at closest approaches (0.64 / 0 / 0.64 / 0).
- Charges and oscillation axis: 7 charge sets × 5 axes. In every case |L| = 0 at every second closest approach.
- Code: the integrator's magnetic force is q v × B (checked numerically); CM momenta balance to 1e-16; d max agrees with energy conservation.
- Signed L: L stays on one axis and touches 0; it never changes sign, so |L| is not hiding a sign flip.
- Torque per half-oscillation: change of L ±0.643 = EM torque ±0.556 + string torque ±0.087; residual < 1e-4.
- String torque at B = 0 (no EM torque): the quirks' L drops to 0.831 = 1/γ* at turning points and returns to 1 at closest approaches, for v_cm = 0, 0.0975 and 0.3. It comes from the v‖v⊥ term of Eq. 3 (the string carries part of L): the same 0.643 when integrated directly in the CM frame.
- Scan: 7 definitions of L, 60 random charge / momentum configurations, 8 equation-of-motion variants: |L(2nd)|/|L(1st)| <= 0.012 (paper 0.30), except forward Euler (0.11 at 300 steps), which does not conserve energy (d max grows 3.6 -> 4.2 cm).

Result

With the paper's Eq. 3 and the Lorentz force, the torque in one half-oscillation is cancelled exactly in the next.
No frame, charge set or axis we tried gives the paper's unequal half-oscillations (|L| 0.19 and 0.29 at the 2nd and 4th approach).
The difference is not explained by anything stated in the paper.
Annihilation: at every second closest approach our |L| is < 1e-5 of its peak (< 6e6 ħ at 100000 steps per period, not converged), i.e. L ≈ 0 where d ≈ 0. The paper states that |L| = 0 only at large separation, which suppresses annihilation. Neither the paper nor this work simulates annihilation.

Questions for the authors:
1. In Fig. 2, in which frame and about which point is L computed; does L include the string's share?
2. What were the event's charges and initial momenta?
3. Did the equation of motion contain anything besides Eq. 3 and the Lorentz force? (The paper and the thesis of J. Setford, Sussex 2018, Ch. 5, state that material effects were neglected; the thesis repeats Fig. 2 without further detail.)

Details: DECISIONS_LOG entries D17-D29; scripts D8, D15-D22 (tables/D15-D22_*.txt).